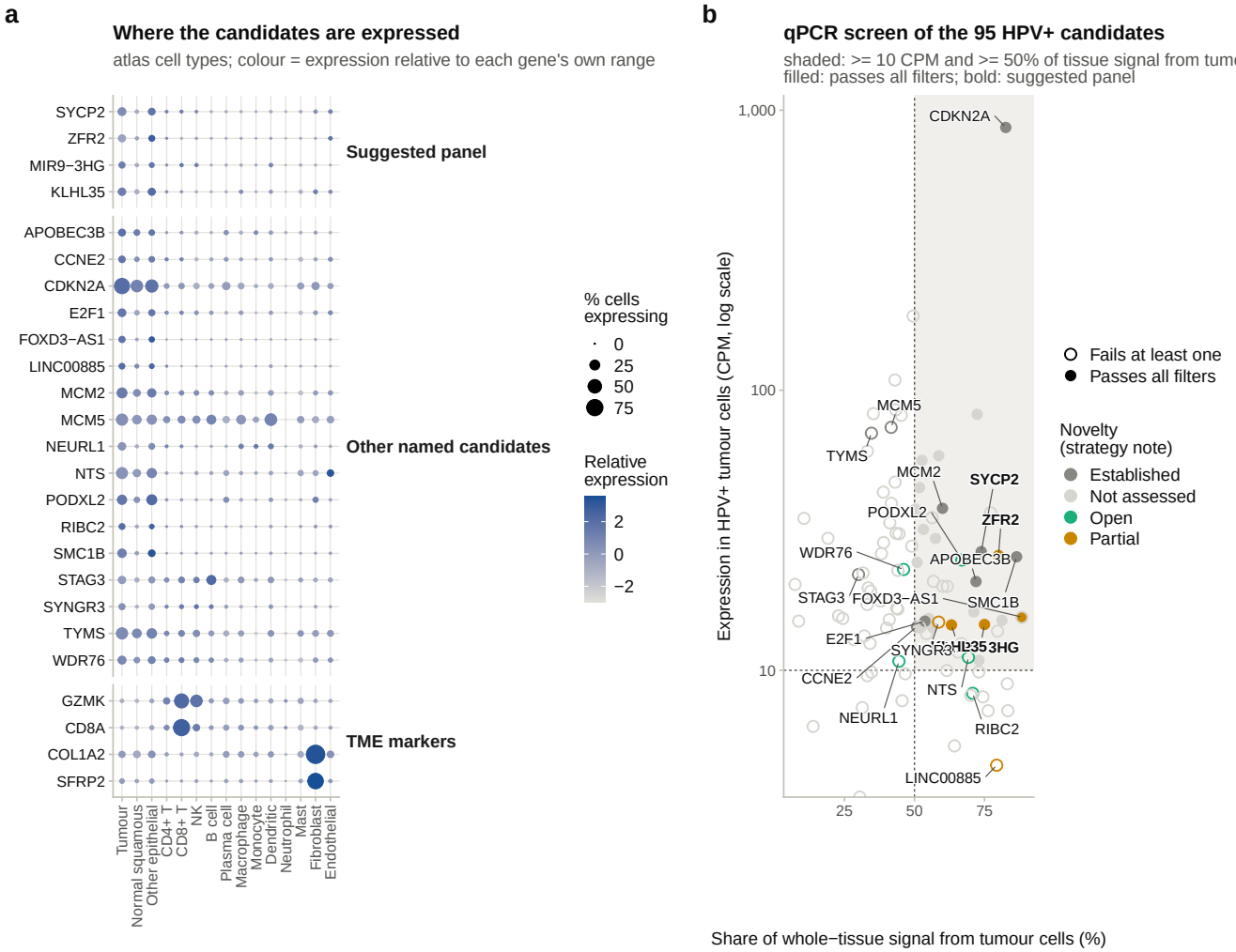
a
Where the candidates are expressed
atlas cell types; colour = expression relative to each gene's own range
SYCP2
ZFR2
MIR9−3HG
KLHL35
Suggested panel
APOBEC3B
CCNE2
CDKN2A
E2F1
FOXD3−AS1
LINC00885
MCM2
MCM5
NEURL1
NTS
PODXL2
RIBC2
SMC1B
STAG3
SYNGR3
TYMS
WDR76
Other named candidates
GZMK
CD8A
COL1A2
SFRP2
TME markers
Tumour
Normal squamous
Other epithelial
CD4+ T
CD8+ T
NK
B cell
Plasma cell
Macrophage
Monocyte
Dendritic
Neutrophil
Mast
Fibroblast
Endothelial
% cells expressing
0
25
50
75
Relative expression
2
0
−2
b
qPCR screen of the 95 HPV+ candidates
shaded: >= 10 CPM and >= 50% of tissue signal from tumour
filled: passes all filters; bold: suggested panel
Expression in HPV+ tumour cells (CPM, log scale)
1,000
100
10
25
50
75
Share of whole−tissue signal from tumour cells (%)
CDKN2A
MCM5
TYMS
MCM2
SYCP2
PODXL2
ZFR2
WDR76
APOBEC3B
STAG3
FOXD3−AS1
SMC1B
E2F1
SYNGR3
HLA-B
35
3HG
CCNE2
NTS
NEURL1
RIBC2
LINC00885
Fails at least one
Passes all filters
Novelty
(strategy note)
Established
Not assessed
Open
Partial